The Recycling Folded Cascode: A General Enhancement of the Folded Cascode Amplifier

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Abstract—**A recycling amplifier architecture based on the folded cascode transconductance amplifier is described. The proposed amplifier delivers an appreciably enhanced performance over that of the conventional folded. This is achieved by using previously idle devices in the signal path, which results in an enhanced transconductance, gain, and slew rate. Moreover, the input referred noise and offset analyses are included to demonstrate that the proposed modifications have no adverse effects on these design metrics. Transistor-level simulations and experimental results in TSMC 0.18 μm CMOS process confirm the theoretical results. When compared to the conventional folded cascode, and for the same area and power budgets, the proposed amplifier has almost twice the bandwidth (134.2 MHz versus 70.7 MHz) and better than twice the slew rate (94.1 V/μs versus 42.1 V/μs) while driving the same 5.6 pF load. Also a gain enhancement of 7.6 dB is observed.**

Index Terms—**Amplifiers, CMOS analog integrated circuits, fast operational amplifiers, low-power low-voltage integrated circuits, operational amplifiers, operational transconductance amplifiers.**

I. Introduction

The advancement of CMOS technologies paved the road for a growing market of mobile and portable electronic devices. This growth is driven by the continual integration of complex analog and digital building blocks on a single chip, making silicon area and power consumption the two most valued aspects of a design. The operational transconductance amplifier (OTA) is still a vital analog building block and for many applications is the largest and most power consuming.

Recently, one of the most commonly used architectures, whether as a single-stage or first stage in multi-stage amplifiers, had been the folded cascode (FC) amplifier for its high gain and reasonably large signal swing in the present and future low voltage CMOS processes. Moreover, the PMOS input FC has become the prime choice over its NMOS counterpart for its higher non-dominant poles, lower flicker noise, and input common mode level. The latter allows input switching using a single NMOS transistor in switched-capacitor (SC) applications [1], [2].

Previous work to enhance the performance of the FC used multi-path schemes [3] and [4]. Another multi-path scheme [5] was applied to the Three-Current-Mirror OTA to enhance the output impedance and slew rate, and another in [6] to emulate a class AB operation. However, they were not suitable for high-speed applications as the transfer function of the OTA had numerous low frequency pole-zero pairs. Nonetheless, [3]–[6] form the basis of the proposed modifications to the FC amplifier presented in Section II. In Section III, the effects of these modifications on the major design metrics of the proposed FC amplifier are presented, whereas Sections IV and V discuss the implementation, and experimental/simulations results respectively. Finally, the conclusions are presented in Section VI.

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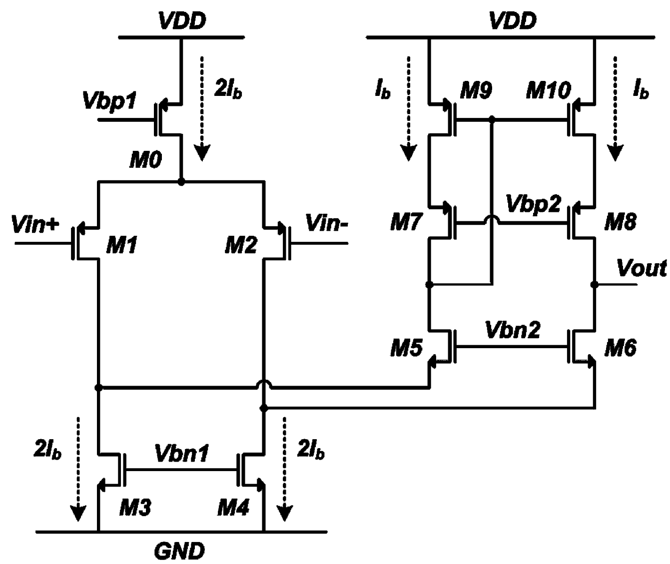

Fig. 1. The conventional folded cascode amplifier.

II. Proposed FC Amplifier

The conventional FC is shown in Fig. 1. Note how transistors $M3$ and $M4$ conduct the most current, and in many designs have the largest transconductance. However, their role is only limited to providing a folding node for the small signal current generated by the input drivers ($M1$ and $M2$).

To address this inefficiency, a modified FC is presented in Fig. 2.The proposed modifications are intended to use $M3$ and $M4$ as driving transistors [7]. First, the input drivers, $M1$ and $M2$ (Fig. 1), are split in half to produce transistors $M1a$, $M1b$, $M2a$, and $M2b$, which now conduct fixed and equal currents of $I_b/2$ (Fig. 2). Next, $M3$ and $M4$ (Fig. 1) are split to form the current mirrors $M3a$:$M3b$, and $M4a$:$M4b$ with a ratio of $K:1$ (Fig. 2). The cross-over connections of these current mirrors ensure the small signal currents added at the sources of $M5$ and $M6$ are in phase. Finally, $M11$ and $M12$ are sized similar to $M5$ and $M6$, and their addition helps maintain the drain potentials of $M3a$:$M3b$ and $M4a$:$M4b$ equal for improved matching.

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